MMWAVE DFP 02.04.18.01 Release Notes

1 Introduction

TI mmWave Device Firmware Package (DFP) enables the development of millimeter wave (mmWave) radar applications using xWR254x/xWR294x/xWR2x4xP/xWR2x4xECO/xWR2x4xLC Device/EVM. It includes necessary components which will facilitate end users to use the RADARSS/RF subsystem in the device. The user is expected to use mmWave mcuplus SDK to integrate with application.

In addition, DFP provides mmWaveLink framework and example application which will serve as a guide for using the mmWave DFP of xWR254x/xWR294x/xWR2x4xP/xWR2x4xECO/xWR2x4xLC

2 Release Overview

2.1 Platform and Device Support

The device and platforms supported with this release include:

Supported Devices	Release Status	Supported EVMs
xWR294xECO ES1.0	Production Release	xWR294xECO EVM
xWR2E4xECO ES1.0	Production Release	xWR2E4xECO EVM
xWR294xP ES1.0	Production Release	xWR294xP EVM
xWR2E4xP ES1.0	Production Release	xWR2E4xP EVM
xWR254x ES1.0	Production Release	xWR254x EVM
xWR294x ES2.0	Production Release	xWR294x EVM
xWR2x4xLC ES1.0	Production Release	xWR2x4xP EVM

Note: DFP supports the foundation components for the device mentioned in the table above. At system level, the mmWave SOC/EVM may interface with other SOCs/EVMs and software for other devices will not be a part of the DFP

2.2 Release contents and component versions

2.2.1 Release contents for xWR2x4xP/xWR2x4xECO/xWR2x4xLC devices

Component	Version	Type
RadarSS Firmware	ROM 2.5.4.0 PATCH: 2.6.3.1	Binary
mmWaveLink Framework	2.4.6.17	Source and Binary
Docs	Release Notes DFP user guide Interface Control Document mmWaveLink Programmer's guide	PDF PDF PDF Doxygen HTML

2.2.2 Release contents for xWR254x devices

Component	Version	Type
RadarSS Firmware	ROM 2.5.4.0 PATCH: 2.5.8.1	Binary
mmWaveLink Framework	2.4.6.17	Source and Binary
Docs	Release Notes DFP user guide Interface Control Document mmWaveLink Programmer's guide	PDF PDF PDF Doxygen HTML

2.2.3 Release contents for xWR294x devices

Component	Version	Type
RadarSS Firmware	ROM 2.4.5.3 PATCH: 2.4.11.0	Binary
mmWaveLink Framework	2.4.6.17	Source and Binary
Docs	Release Notes DFP user guide Interface Control Document	PDF PDF PDF

	mmWaveLink Programmer's guide	Doxygen HTML

2.3 Directory Structure

Directory Name	Content
Docs	mmWave-Radar-Interface-Control.pdf mmwave_dfp_release_notes.pdf mmwave_dfp_user_guide.pdf
Firmware	RadarSS firmware binary files in RPRC format xwr2x4xp_radarss_metarprc.bin xwr25xx_radarss_metarprc.bin xwr29xx_radarss_metarprc.bin mmwave_radarss_xWR2x4xP_release_notes.pdf mmwave_radarss_xWR254x_release_notes.pdf mmwave_radarss_xWR294x_release_notes.pdf
Ti	mmWaveLink framework mmWaveLink Programmer's guide
rf_eval	RF evaluation firmware

2.4 Component Descriptions

2.4.1 RadarSS Firmware

Refer to Radar SS firmware release notes (mmwave_radarss_[device]_release_notes.pdf) in firmware**radarss** folder.

2.4.2 mmWaveLink framework

RadarSS is a closed subsystem whose internal blocks are configurable using messages coming over mailbox.
TI mmWaveLink framework acts as driver for RadarSS, and exposes services of mmWave Radar. It includes APIs to configure HW blocks of RadarSS and provides communication protocol for message transfer between the application and the RadarSS.

- Link between application and RadarSS
- Platform and OS independent which means it can be ported into any processor which provides communication interface such as SPI and basic OS routines. The mmWaveLink framework can also run in single threaded environment
- The LDRA tool v9.7.6 with MISRA 2012 AMD1 rule is used to perform static analysis; the TI approved waiver policy is being used to take any waiver.

2.5 Licensing

Please refer to the mmwave_dfp_manifest.html, which outlines the licensing information for mmWave DFP package.

3 Release Contents

3.1 xWR294x ES1.0 RAM contents.

3.1.1 DFP 2.4.0.1 Features

- xWR294x is TIs 77GHz RF CMOS Radar SOC. Features supported in this firmware release:
    - This release is derived from AWR2243 DFP2.2 release baseline and backward compatible to DFP 2.2 supported APIs with minor updates for enhancement.
    - Supports 4 TX and 4 Real-only RX chains.
    - Synthesizer RF frequency supported 76 – 81GHz
        - VCO1: 76 – 77GHz
        - VCO2: 76 – 81GHz
    - Supports 7 functional profiles.
    - Supports 266MHz/us max slope
    - Supports programmable filter (not characterized in this engineering release)
- Calibrations
    - APLL calibration.
    - SYNTH calibration.
    - Other calibrations are **NOT** supported in this wakeup release.
- Monitoring
    - Digital / Analog-RF monitors are **NOT** supported in this wakeup release.
- Refer mmWave-Radar-Interface-Control.pdf (ICD) for more information on xWR294x ES1.0 API details.

3.1.2 DFP 2.4.0.2 Features and Enhancements.

- Support for other boot and run-time calibrations
    - LO-DIST calibration.
    - ADC-DC calibration.
    - RX HPF calibration.

  - RX LPF calibration.
  - PD calibration.
  - TX power calibration.
  - RX gain calibration.
  - TX phase shifter calibration.
- Support for calibration data save/restore feature.
- Monitoring
  - Digital / Analog-RF monitors are **NOT** supported in this release.

3.1.3 DFP 2.4.1.1 Features and Enhancements.

- Support for Digital boot and periodic monitors.
- Support for Analog/RF periodic monitors.
- Support for runtime TX phase shifter calibration.
- Support for new VMON configuration API.
- Improved IFA bias settings for low power ADC modes.
- New Bus safety and ECC aggregator boot monitors.

3.1.4 DFP 2.4.2.1 Features and Enhancements.

- Support for advanced chirp configuration APIs.
- MPU digital boot monitor added.
- New API to add override and dither configurations for RF monitors.
- Improvements in boot time TX power calibration.
- Improvements in RX gain calibration.
- New and optimized TX PS DAC monitor.
- New inter-burst power save options (APLL, LDO duty cycling).
- Improvements in the RF PVT LUTs for power saving.
- Optimizations in TX monitors for power saving.

3.1.5 DFP 2.4.2.2 Features and Enhancements.

- No new features / enhancements when compared to DFP 2.4.2.1.

3.1.6 DFP 2.4.3.1 Features and Enhancements.

- Support for dynamic per-chirp phase shifter configuration API for 4TX.

- Support for enabling the RSS logger through an API field.
- Improved SYNTH performance by VCO specific configurations.
- Improvements to the TX power calibration accuracy.
- Support for 3 new loopback frequencies in RF loopback.
- Improved TX output power performance at the higher end of the RF BW (81GHz).
- Support for API based control of the SYNTH fast reset end time within the chirp idle time.
- Accuracy improvements in boot and periodic TX phase shifter calibration.
- Blocking the below unsupported periodic analog monitors in AWR_MONITOR_ANALOG_ENABLES_CONF_SB API.
    - RX noise monitor.
    - External analog signals monitor.
    - RX mixer power monitor.
    - RX signal and image band monitor.

3.1.7 DFP 2.4.5.0 Features and Enhancements.

- Improvements to SYNTH frequency error monitoring.
- Improved APLL/SYNTH BW settings.
- BSS logger data can now be sent to a configurable MSS memory buffer from which it can be transmitted out through peripherals like UART/Ethernet.
- Support for digital TX frequency shifting even when loopback is enabled.
- New fault injection sequences for supply LDO and VCO open loop faults.
- Improved error response to VMON self-test failures.
- Improved TX PS boot time calibration.
- New TX loopback sequence updates for ES2.0.
- New RX IF settings for ES2.0
- Temperature based VCO LDO output voltage settings.

3.2 xWR294x ES2.0 Patch contents

xWR294x ES2.0 has DFP 2.4.5.0 contents in ROM and the below DFP releases are applicable only for ES2.0 through firmware patch updates.

3.2.1 DFP 2.4.6.0 Features and Enhancements.

- Improved temperature sensor accuracy by using an EFUSE based two point trimming process.
- Improved RF loopback isolation leading to better calibration and monitoring performance in ES2.0 devices.

- Improved inter-RX and inter-TX mismatches on ES2.0 devices.
- Improved VCO LDO settings for ES2.0 devices.
- Improved RX gain accuracy for ES2.0 devices.
- Improved TX gamma measurement accuracy for ES2.0 devices.

3.2.2 DFP 2.4.8.1 Features and Enhancements.

- Improved RX RF gain targets for better RX gain accuracy on ES2.0 devices.
- Improved TX power monitoring stability.

3.2.3 DFP 2.4.9.1 Features and Enhancements

- Improved timings for periodic TX phase shifter calibration.
- Improved temperature sensor accuracy.

3.2.4 DFP 2.4.14.0 Features and Enhancements

- Support for multiple profiles in synth frequency monitor (Live Mode).
- Improved minimum chirp cycle time for advanced chirp configuration by allowing enable/disable control for parameters in advanced chirp configuration.

3.2.5 DFP 2.4.17.0 Features and Enhancements

- No changes to patch binary from DFP 2.4.14.0
- mmWaveLink Framework version updated
- Defeatured OSCLKOUT in xWR294x devices. Refer mmwave_radarss_xWR294x_release_notes.doc for RadarSS documentation update details

3.2.6 DFP 2.4.18.0 Features and Enhancements

- Fixes for Group1 ECC Aggregator SEC and DED error

3.2.7 DFP 2.4.18.1 Features and Enhancements

- No change to patch binary from DFP 2.4.18.0
- Defeatured RX FE disabled mode for loopback burst configurations. Refer mmwave_radarss_xWR294x_release_notes.doc for RadarSS documentation update details

3.3 xWR254x ES1.0 Patch contents

xWR254x ES1.0 has RadarSS Firmware contents in ROM and the below DFP releases provide patch on top of ROM. The below DFP release are applicable only for ES1.0 through firmware patch updates.

3.3.1 DFP 2.4.11.0 Features and Enhancements

- xWR254x is TIs 77GHz RF CMOS Radar SOC. Features supported in this firmware release
  - xWR254x RadarSS ROM Firmware is derived from AWR2944 DFP2.4.9 release baseline with additional enhancements. It is backward compatible to DFP 2.4.9 supported APIs with minor updates for enhancement.
  - Supports 4 TX and 4 Real-only RX chains
  - Synthesizer RF frequency supported 76 – 81GHz
  - VCO1: 76 – 77GHz
  - VCO2: 76 – 81GHz
  - Supports 7 functional profiles.
  - Supports 266MHz/us max slope
  - Supports programmable filter
  - Support for inter-chirp jitter mitigation
  - Synth frequency monitor (Live Mode) support for multiple profiles
  - Support for enabling 1 ns resolution for ADC start time in advanced chirp configuration
  - Added new FRC DCC clock monitor
  - Added support to disable parameters in advanced chirp configuration
  - Added support to enable/disable OSCCLKOUTETH to provide reference clock for external ethernet PHY
- Refer mmWave-Radar-Interface-Control.pdf (ICD) for more information on xWR254x ES1.0 API details.

3.3.2 DFP 2.4.12.0 Features and Enhancements

- Improvements in LODIST calibration
- Fix for monitoring time stamps when RSS dynamic frequency switch is enabled and working with 50MHz XTAL

3.3.3 DFP 2.4.12.1 Features and Enhancements

- mmWaveLink examples are moved from DFP mmwave_dfp_[ver]\ti\control\mmwavelink\test to mmwave_mcuplus_sdk_[ver]\mmwave_mcuplus_sdk_[ver]\ti\control\mmwave\link_test
- xWR254x : RadarSS Firmware and mmWaveLink same as DFP 2.4.12.0

- xWR294x : RadarSS Firmware same as DFP 2.4.9.1. mmWaveLink updated for xWR254x device specific changes. Refer section 3.5.3, 3.5.4 and 3.5.5 for details

3.3.4 DFP 2.4.13.0 Features and Enhancements

- Adjusted synth control voltage monitor and fault injection for robustness
- Enabled debug support for LPF monitor in RX IFA Stage monitor
- Improved operation for inter-chirp jitter mitigation mode
- Improved temperature sensor accuracy.

3.3.5 DFP 2.4.14.0 Features and Enhancements

- Minor bug fixes

3.3.6 DFP 2.4.17.0 Features and Enhancements

- No changes to patch binary from DFP 2.4.14.0
- mmWaveLink Framework version updated
- Defeatured OSCLKOUT in xWR294x devices. Refer mmwave_radarss_xWR254x_release_notes.doc for RadarSS documentation update details

3.3.7 DFP 2.4.18.0 Features and Enhancements

- No changes from DFP 2.4.17.0

3.3.8 DFP 2.4.18.1 Features and Enhancements

- No changes from DFP 2.4.18.0

3.4 xWR2x4xP ES1.0 Patch contents

xWR2x4xP ES1.0 has RadarSS Firmware contents in ROM and the below DFP releases provide patch on top of ROM. The below DFP release are applicable only for ES1.0 through firmware patch updates.

3.4.1 DFP 2.4.15.0 Features and Enhancements

- xWR2x4xP is TIs 77GHz RF CMOS Radar SOC. Features supported in this firmware release
    - xWR2x4xP RadarSS ROM Firmware and patch Firmware is derived from xWR294x/xWR254x DFP2.4 release baseline. It is backward compatible to DFP 2.4 supported APIs with minor updates for enhancement.
    - Supports 4 TX and 4 Real-only RX chains

  - Synthesizer RF frequency supported 76 – 81GHz
  - VCO1: 76 – 77GHz
  - VCO2: 76 – 81GHz
  - Supports 7 functional profiles.
  - Supports 266MHz/us max slope
  - Supports programmable filter
  - Support for inter-chirp jitter mitigation
  - Synth frequency monitor (Live Mode) support for multiple profiles
  - Support for enabling 1 ns resolution for ADC start time in advanced chirp configuration
  - Supports FRC DCC clock monitor
  - Improved minimum chirp cycle time for advanced chirp configuration by allowing enable/disable control for parameters in advanced chirp configuration
  - Support to enable/disable OSCCLKOUTETH to provide reference clock for external ethernet PHY
  - Support for 20Mhz IF bandwidth
  - Improved TX Power
  - Improved RX Noise Figure

- Refer mmWave-Radar-Interface-Control.pdf (ICD) for more information on xWR2x4xP ES1.0 API details.

3.4.2 DFP 2.4.16.0 Features and Enhancements

- Updates to RF Gain Target and Max RX Gain
- Improved TX Power accuracy

3.4.3 DFP 2.4.17.0 Features and Enhancements

- Improved TX Phase shifter accuracy
- Improved TX Power accuracy

3.4.4 DFP 2.4.18.0 Features and Enhancements

- Fixes for Group1 ECC Aggregator SEC and DED error
- Improved temperature sensor accuracy

3.4.5 DFP 2.4.18.1 Features and Enhancements

- No change to patch binary from DFP 2.4.18.0
- Defeatured RX FE disabled mode for loopback burst configurations. Refer mmwave_radarss_xWR2x4xP_release_notes.doc for RadarSS documentation update details

3.5 Feature/Changes List by Components

3.5.1 RadarSS firmware

Refer to RadarSS firmware release notes (mmwave_radarss_xWR2xxxx_release_notes.pdf) in firmware**radarss** folder

3.5.2 mmWaveLink framework (DFP 2.4.8.1 / DFP 2.4.9.1/AWR294x compared to DFP 2.2/AWR2243)

Type	Key	Description
Enhancement	MMWL-350 MMWL-351 MMWL-353 MMWL-355	Initial porting of xWR294x specific API updates to the mmWaveLink framework.
Enhancement	MMWL-356	Porting of AWR2243 maintenance release DFP 2.2.3 changes to the xWR294x mmWaveLink. 1. Adding frequency dither limits in RX gain phase monitoring API. 2. Added new fields in APLL SYNTH BW control API and some doxygen comment updates. 3. Fixed error in byte ordering for TX gain phase offset config compared to the ICD. 4. Fixed a few mmWaveLink compiler warnings. 5. Updated Die-ID Get API fields and doxygen comments.
Enhancement	MMWL-358	Added 4th TX monitoring API updates.
Enhancement	MMWL-363	1. De-featured the existing TX PS DAC monitoring controls and reports (available in Internal TX signals monitor) and added a new configuration and async report for TX phase shifter DAC monitor (rlRfMontxPhShiftDacConfig). 2. Added a new API for configuring the RF parameter overrides for monitoring chirps (rlRfMonOvrdDitherConfig). 3. Added a bit for controlling the inter-burst APLL power save feature in the rlRfDevCfg_t structure.

Enhancement	MMWL-364	1. De-featured rlRfDfeRxStatisticsReport and added a new API (rlRfRealChDfeRxStatisticsReport) for xWR294x. 2. Added a new field to control CRD slope magnitude in the rlRfSetMiscConfig API.
Enhancement	MMWL-366	1. De-featured rlRfTxGainPhaseMismatchMonConfig and added a new API (rlRfAdvTxGainPhaseMismatchMonConfig) for xWR294x. 2. De-featured rlSetDynPerChirpPhShifterCfg and added a new API (rlSetAdvDynPerChirpPhShifterCfg) for xWR294x. 3. A few miscellaneous doxygen comment updates.
Enhancement	MMWL-368	1. Added controls for configuring the synthesizer fast reset pulse's end time during the chirp idle time in the rlRfSetMiscConfig API. 2. De-featured rlSetGpAdcConfig API. 3. Updated the phase value mapping in rlRfPhShiftCalibDataRestore and rlRfPhShiftCalibDataStore APIs.
Enhancement	MMWL-367	mmWaveLink library generation moved to TI CLANG compiler.
Enhancement	MMWL-372	1. Updated the rlRfApllSynthBwCtlConfig to take synthesizer ICP and RZ trims for each VCO independently. 2. Doxygen comment updates.
Bug	MMWL-373	Fixed a 4-byte alignment issue in the monitoring report structure.
Bug	MMWL-378	Fixed MISRA C and static analysis violations.
Bug	MMWL-379	Fixed bugs identified during code inspection. 1. Correct value of macro for specifying number of profiles in rlRealChDfeStatReport_t 2. Fix big endian implementation for rlAdvChirpPhShiftPerTx structure 3. Fix big endian implementation for field RF_TX_NOISE_POWER in rlMonAdvTxGainPhaMisRep_t 4. Remove hard coded value used for num of Tx channels for rlRfPhShiftCalibDataRestore, rlRfPhShiftCalibDataStore functions 5. Use macros instead of hard coded values in rlMonOvrdDitherConf_t structure 6. Use macros instead of hard coded values in rlMonAdvTxGainPhaMisRep_t structure 7. Update Rx gain values in code documentation if there is any mismatch between ICD and code

Enhancement	MMWL-382	1. De-featured rlRfGetEsmFault and added a new API rlRfGetAdvEsmFault for xWR294x. 2. De-featured LPF monitoring related controls and results in the rlRfRxIfStageMonConfig and its async monitoring report. 3. Added error codes related to the new APIs added in xWR294x.
Enhancement	MMWL-388	1. Updated rlRfSetCalMonFreqLimitConfig and rlRfTxFreqPwrLimitConfig with a new vco2RangeConfig field. 2. Doxygen comment updates.
Bug	MMWL-383	1. Fixed the issue of missing 4th TX configuration in the advanced chirp config LUT load API. 2. Fixed the issue of CHIRP_TX_EN and CHIRP_BPM_EN fields in advanced chirp configuration APIs not allowing the 4th TX to be configured.
Enhancement	MMWAVE_DFP-1838	Updated the default VCO2 range to 76.5 to 81GHz.
Enhancement	MMWAVE_DFP-1840	Added a device MACRO for xWR294x and some doxygen comment updates.
Enhancement	MMWAVE_DFP-1850	Updated doxygen comments for the new RF gain targets in rlSetProfileConfig API.
Enhancement	MMWL-393	Doxygen comment updates to take in the ICD changes.
Bug	MMWL-394	Fixing MISRA and static analysis violations.

3.5.3 mmWaveLink framework (DFP 2.4.11.0/xWR254x compared to DFP 2.4.9/xWR294x)

Changes specified DFP2.4.11.0 onwards in all the tables below table are applicable for all devices unless device is specified

Type	Key	Description
Enhancement	MMWL-401	**Updates to support xWR254x devices:** Added support for 1ns resolution in ADC Start time for advanced chirps and added support for inter chirp jitter mitigation for xWR254x devices Updated AWR_RF_RADAR_MISC_CTL_SB (rlRfSetMiscConfig) 1. Bit 6 (ADC_START_TIME_RES) added in “miscCtl” field to support 1 ns resolution of ADC start time. 2. Bit 7 (INTER_CHIRP_JITTER_MITIGATION) added in field”miscCtl” to enable/disable features to mitigate jitter in synchronization of sequencer clock and synthesizer clock, resulting in improved phase stability across chirps.

		Added support for FRC DCC monitor for xWR254x devices Updated AWR_MONITOR_DUAL_CLOCK_COMP_CONF_SB (rlRfDualClkCompMonConfig) 1. Bit 6 (FRC_200M) added in “dccPairEnables” field to enable FRC clock monitoring. Updated AWR_MONITOR_DUAL_CLOCK_COMP_REPORT_AE_SB (rlMonDccClkFreqRep_t) 1. Bit 6 (STATUS_CLK_PAIR6) added in “statusFlags” fields to report FRC clock frequency monitoring status if enabled. 2. Byte 12, 13 (FRC_200M) added in “freqMeasVal” field to report FRC clock frequency if it is enabled for monitoring. Added support for OSCCLKOUTETH for xWR254x devices Updated AWR_CHAN_CONF_SET_SB (rlSetChannelConfig) 1. Bit 7 (OSCCLKOUTETH_EN) added in “cascadingPinoutCfg” field to enable/disable OSCCLKOUTETH 2. Bit 8 (OSCCLKDIV_BIT) added in “cascadingPinoutCfg” field to enable/disable divider for OSC_CLKOUT/OSC_CLKOUT_ETH 3. Bits [12:9] (OSCLKOUTETH_DRV_VAL) added in “cascadingPinoutCfg” field to configure drive strength of OSC_CLKOUT_ETH Defeatured BIST FFT self test , Bus safety self test, DFE And Parity for xWR254x devices Updated AWR_RF_BOOTUPBIST_STATUS_GET_SB (rlGetRfBootupStatus) 1. Bit 24 (FFT test) in “bssSysStatus” field has been marked as reserved as BIST FFT is defeatured 2. Bit 27 (Bus Safety test) in “bssSysStatus” field has been marked as reserved as bus safety self-test is defeatured Updated AWR_MONITOR_RF_DIG_LATENTFAULT_REPORT_AE_SB (rlDigLatentFaultReportData_t) 1. Bit 24 (FFT test) in “digMonLatentFault” field has been marked as reserved as BIST FFT is defeatured Updated AWR_MONITOR_RF_DIG_LATENTFAULT_CONF_SB (rlRfDigMonEnableConfig)

		1. Bit 24 (FFT test) in “enMask” field has been marked as reserved as BIST FFT is defeatured Updated AWR_RF_ADV_ESMFAULT_STATUS_GET_SB (rlRfGetAdvEsmFault) 1. Bit 4 (DFE_PARITY_AND_OUT_ERROR) in “esmGrp1ErrLsb” field has been marked as reserved as DFE_PARITY_AND is defeatured. 2. Bit 30 (BUS_SAFETY_PCR_ERROR), Bit 31 (BUS_SAFETY_BSS_SLV_ERROR) in “esmGrp1ErrLsb” field has been marked as reserved as bus safety self-test is defeatured 3. Bit 0 (BUS_SAFETY_BSS_MST_ERROR), Bit 1 (BUS_SAFETY_STATIC_MEM_ERROR), Bit 2 (BUS_SAFETY_MBOX_ERROR), Bit 3 (BUS_SAFETY_SEC_AGGREGATED_ERROR) in “esmGrp1ErrMsb” field has been marked as reserved as bus safety self-test is defeatured. Updated AWR_AE_RF_ADV_ESMFAULT_SB (rlBssAdvEsmFault_t) 1. Bit 4 (DFE_PARITY_AND_OUT_ERROR) in “esmGrp1ErrLsb” field has been marked as reserved as DFE_PARITY_AND is defeatured. 2. Bit 30 (BUS_SAFETY_PCR_ERROR), Bit 31 (BUS_SAFETY_BSS_SLV_ERROR) in “esmGrp1ErrLsb” field has been marked as reserved as bus safety self-test is defeatured. 3. Bit 0 (BUS_SAFETY_BSS_MST_ERROR), Bit 1 (BUS_SAFETY_STATIC_MEM_ERROR), Bit 2 (BUS_SAFETY_MBOX_ERROR), Bit 3 (BUS_SAFETY_SEC_AGGREGATED_ERROR) in “esmGrp1ErrMsb” field has been marked as reserved as bus safety self-test is defeatured. Defeature Digital RX Gain compensation for xWR254x devices Updated AWR_DIGITAL_COMP_EST_CONTROL_SB (rlRfInterRxGainPhaseConfig) 1. Bit 0 (Digital RX gain compensation enable) in “digCompEn” field has been marked as reserved as digital RX gain compensation is defeatured. 2. Field “digRxGainComp” has been marked as reserved as digital RX gain compensation is defeatured. Added support for multiple profiles for synthesizer frequency monitor (live) for xWR254x devices Updated

		AWR_MONITOR_SYNTHESIZER_FREQUENCY_CONF_SB (rlRfSynthFreqMonConfig) 1. Added a new field “profileBitMaskLiveMode” that allows user to enable/disable monitoring for multiple profiles in live mode. 2. Either the legacy field “profileIndx” or the new field “profileBitMaskLiveMode” can be used at a time. It is not recommended to use both fields together. Added disable field to control enable/disable of a parameter for advance chirp configuration for xWR254x devices Updated AWR_ADVANCE_CHIRP_CONF_SB (rlSetAdvChirpConfig) 1. New field “paramControl” added to enable/disable parameter control.

3.5.4 mmWaveLink framework (DFP 2.4.12.0 compared to DFP 2.4.11.0)

Type	Key	Description
Enhancement	MMWL-407	Doxygen comment updates to take in the ICD changes: 1. Added note in miscCtl of rlRfMiscConf. Recommendation to enable CRD when inter chirp jitter mitigation is enabled for xWR254x 2. Updated BUS_SAFETY_BSS_MST_ERROR, BUS_SAFETY_STATIC_MEM_ERROR, BUS_SAFETY_MBOX_ERROR, BUS_SAFETY_SEC_AGGREGATED_ERROR as reserved for xWR254x in esmGrp1ErrMsb field of rlBssAdvEsmFault

3.5.5 mmWaveLink framework (DFP 2.4.12.1 compared to DFP 2.4.12.0)

No change from DFP 2.4.12.0

3.5.6 mmWaveLink framework (DFP 2.4.13.0 compared to DFP 2.4.12.1)

Type	Key	Description
Enhancement	MMWL-408	Doxygen comment updates to take in the ICD changes: 1. Doxygen comment updates for updating minimum RX Gain from 24dB to 28dB for xWR254x devices. It is recommended to update the RX Gain settings in these API’s rlGetProfileConfig, rlSetContModeConfig and rlSetLoopBckBurstCfg to the new minimum setting 2. Added note in ps2PgaIndx field of rlLoopbackBurst.

		Recommendation is to keep LO loopback buffer setting lesser than -7dB to avoid ADC saturation. 3. Added note related to inter-chirp jitter mitigation in rlSetFrameConfig and rlSetAdvFrameConfig for xWR254x 4. Added note in rlRfGetTemperatureReport. Recommendation to use AWR_MONITOR_TEMPERATURE_CONF_SB to read periodic temperature during functional frames instead of AWR_RF_TEMPERATURE_GET_SB 5. Updated note in rlRfMiscConf for 1ns ADC start time resolution mode, interchirp jitter mitigation for xWR254x 6. Updated notes (reworded) in b2 “Digital RX delay compensation enable“ of rlInterRxGainPhConf for xWR254x 7. Updated notes (reworded) in rlAdvChirpCfg 8. Added Minimum Chirp Cycle time details for xWR254x devices 9. Added note in lpfCutoffBandEdgeDroopThresh and lpfCutoffStopBandAttenThresh fields of rlRxIfStageMonConf. In xWR254x devices/xWR2x4xP devices LPF monitor is used for debug purposes only. LPF monitor will be executed only when lpfCutoffBandEdgeDroopThresh and lpfCutoffStopBandAttenThresh both are set to non-zero threshold values. Refer the monitoring timing table for LPF monitor enabled/disabled case. 10. Added note related to synth control voltage monitor in rlAnaFaultInj 11. Added Monitoring time details for xWR254x devices.

3.5.7 mmWaveLink framework (DFP 2.4.14.0 compared to DFP 2.4.13.0)

Type	Key	Description
Bug	MMWL-400	Doxygen comment updates for updating minimum RX Gain from 24dB to 28dB for xWR294x devices. It is recommended to update the RX Gain settings in these API’s rlGetProfileConfig, rlSetContModeConfig and rlSetLoopBckBurstCfg to the new minimum setting.
Enhancement	MMWL-399	Doxygen comment updates to take in the ICD changes for xWR254x and xWR294x devices: Added support for multiple profiles for synthesizer frequency monitor (live) in xWR294x devices

		Updated AWR_MONITOR_SYNTHESIZER_FREQUENCY_CONF_SB (rlRfSynthFreqMonConfig) 1. Field “profileBitMaskLiveMode” that allows user to enable/disable monitoring for multiple profiles in live mode is applicable for xWR294x devices 2. Either the legacy field “profileIndx” or the new field “profileBitMaskLiveMode” can be used at a time. It is not recommended to use both fields together. Added disable field to control enable/disable of a parameter for advance chirp configuration in xWR294x devices 1. In AWR_ADVANCE_CHIRP_CONF_SB (rlSetAdvChirpConfig) field “paramControl” is applicable for xWR294x devices Updated note related to synth control voltage monitor in rlAnaFaultInj
Bug	MMWL-410	Update to fields rxGainMismatchOffsetVal and rxGainPhaseMismatchOffsetVal of rlRfRxGainPhMonConfig API. Fields are swapped to fix bug.
Bug	MMWL-411	Datatype update from unsigned to signed to support negative values for these fields rfFreqSlopeOvrd, rfMinDitherFreq, rfMaxDitherFreq in rlRfMonOvrdDitherConfig API

3.5.8 mmWaveLink framework (DFP 2.4.15.0 compared to DFP 2.4.14.0)

All mmWaveLink framework updates for xWR254x devices are applicable for xWR2x4xP devices

Type	Key	Description
Enhancement	MMWL-412	Added support for xWR2x4xP devices

3.5.9 mmWaveLink framework (DFP 2.4.16.0 compared to DFP 2.4.15.0)

Type	Key	Description
Enhancement	MMWL-414	Doxygen comment updates: 1. Updating minimum RX Gain from 28dB to 30dB for xWR2x4xP devices. It is recommended to update the RX Gain settings in these API’s rlGetProfileConfig, rlSetContModeConfig and rlSetLoopBckBurstCfg to the new minimum setting.

		2. Updating RF Gain Target from 32,34,36 dB to 34, 36, 38 dB for xWR2x4xP device in rlGetProfileConfig, rlSetContModeConfig and rlSetLoopBckBurstCfg

3.5.10 mmWaveLink framework (DFP 2.4.17.0 compared to DFP 2.4.16.0)

Type	Key	Description
Bug	MMWL-409	Fixed support for 7 profiles in rlSetProfileConfig
Bug	MMWL-415	Added a macro RL_RF_AE_ADV_ESMFAULT_SB for RL_RF_ASYNC_EVENT_SB
Enhancement	MMWL-416	Doxygen comment updates to take in ICD changes: 1. Updated comments for xWR2x4xP device 2. Updated note in rlProfileCfg , rlRfTxPowrMonConfig rlRfTxnPowrMonConfig. 0dB backoff corresponds to typically 13.5dBm power level for xWR2x4xP devices. 3. Updated note for OSCCLKOUT_DIS in xWR294x/xWR254x/xWR294xP devices. It is recommended to keep OSCCLKOUT disabled in rlChanCfg 4. Updates to monitoring API description RF1 renamed from Lowest RF Frequency to Starting RF Frequency, RF3 renamed from Highest RF Frequency to End RF Frequency

3.5.11 mmWaveLink framework (DFP 2.4.18.1 compared to DFP 2.4.17.0)

Type	Key	Description
Enhancement	MMWL-418	Doxygen comment updates to take in ICD changes for xWR2x4xECO and xWR2x4xP devices.

4 Unsupported features and APIs (applicable to all DFP releases)

Refer to RadarSS firmware release notes in firmware**radarss** folder for radarSS APIs.

5 Known Issues (applicable to all DFP releases)

5.1 RadarSS firmware known Issues:

Refer to RadarSS firmware release notes (mmwave_radarss_release_notes.pdf) in firmware**radarss** folder.

6 Migration guide

6.1 Migration from AWR2243 (DFP 2.2) to AWR294x (DFP 2.4.9)

This section explains the steps to migrate from AWR2243 (DFP 2.2) release to DFP 2.4.9 package.

Impact	Change list
MEDIUM	Added a new command message ID “RL_RF_MONITORING_CONF_SET_2_MSG” to support new monitoring configuration APIs.
HIGH	Added the following new API messages to support 4th TX added in xWR294x devices. • RL_RF_ADV_TX_GAIN_TEMPLUT_SET_SB rlAdvTxGainTempLutSet rlAdvTxGainTempLutGet • RL_RF_ADV_DYN_PERCHIRP_PHSHFT_CFG_SET_SB rlSetAdvDynPerChirpPhShifterCfg • RL_RF_TXN_POW_MON_CONF_SB rlRfTxnPowrMonConfig • RL_RF_TXN_BALLBREAK_MON_CONF_SB rlRfTxnBallbreakMonConfig • RL_RF_TXN_INT_ANA_SIGNALS_MON_CONF_SB rlRfTxnIntAnaSignalsMonConfig • RL_RF_TXN_PH_SHIFT_MON_CONF_SB rlRfTxnPhShiftMonConfig • RL_RF_ADV_TX_GAIN_PHASE_MIS_MON_CONF_SB rlRfAdvTxGainPhaseMismatchMonConfig • RL_RF_MON_TX_PHSHIFTER_DAC_CONF_SB rlRfMontxPhShiftDacConfig
HIGH	Deprecated RL_RF_RFESMFAULT_STATUS_SB (rlRfGetAdvEsmFault) and added a new API - RL_RF_RFADVESMFAULT_STATUS_SB for supporting additional ESM fault indication bits available in xWR294x.
MEDIUM	Added a new API - RL_RF_VMON_MON_CONF_SB (rlRfVmonMonConfig) to configure and enable the BSS VMONs in xWR294x. Even though these VMONs are configured using RSS APIs, the response to a VMON event (under/over voltage) is directly provided to the

	MSS ESM and should be taken care of by the application.
MEDIUM	Added a new API - RL_RF_MON_OVRD_DITHER_CONF_SB (rlRfMonOvrdDitherConfig) to provide more control over the RF parameters and dither options to be used in different RF/Analog monitoring sequences.
LOW	Added a new API - RL_RF_LOLOOPBACK_CFG_SET_SB (rlRfSetLOLoopbackConfig) for configuration and enabling the new LO loopback circuits.
LOW	Deprecated the PS loopback API - RL_RF_PSLOOPBACK_CFG_SET_SB as it is no longer relevant in xWR294x devices. Customers can use the LO loopback instead.
HIGH	Added the below new ASYNC monitoring reports to accommodate the 4th TX added in xWR294x devices. • RL_RF_AE_MON_TX3_POWER_REPORT • RL_RF_AE_MON_TX3_BALLBREAK_REPORT • RL_RF_AE_MON_TX3_INT_ANA_SIG_REPORT • RL_RF_AE_MON_TX3_PH_SHIFT_REPORT • RL_RF_AE_MON_ADV_TX_GAIN_PHASE_MIS_REPORT • RL_RF_AE_MON_ADV_TX_PHSHIFTER_DAC_REPORT
LOW	Defeatured RL_RF_DFE_STATISTICS_REPORT_GET_SB and added a new API - RL_RF_REAL_CH_DFE_STATISTICS_REPORT_GET_SB (rlRfRealChDfeRxStatisticsReport) to get DFE statistics on xWR294x devices.
MEDIUM	FRAME_TIMING_MONITORING (BIT3) of the "periodicEnableMask" field in the RL_RF_DIG_MON_PERIODIC_CONF_SB API (rlDigMonPeriodicConf_t) cannot be used when RSS dynamic frequency switching is enabled. The customer application can use alternate methods to monitor the frame timing. 1. Periodic monitoring of the FTTI using the MSS PMU to timestamp the temperature monitoring ASYNC reports and computing the period between them. 2. Another option is periodic monitoring of the FRAME_START (burst start) interrupt coming to the MSS. Measuring the time period between two FRAME_START interrupts using the MSS PMU, we should get a more accurate estimate of the frame timing. However, this means that when the customer uses complicated waveforms like advanced frame configuration, they would have to compute the expected value of the current burst period (including IDLE periods), before comparing against the measured FRAME_START -> FRAME_START time. Please refer to the TRM for more details on the FRAME_START interrupt.
HIGH	RL_RF_ANA_MON_EN_SB API (rlMonAnaEnables_t) API has been modified in xWR294x with the below changes. 1. Defeatured the below monitors in "enMask1" field. a) RX_NOISE_MONITOR_EN (Bit 2) b) EXTERNAL_ANALOG_SIGNALS_MONITOR_EN (Bit 15) c) RX_SIG_IMG_BAND_MONITORING_EN (Bit 25) d) RX_MIXER_INPUT_POWER_MONITOR (Bit 26) 2. Added the below new monitors in "enMask1" field.

	a) TX3_POWER_MONITOR_EN (Bit 29) b) TX3_BALLBREAK_MONITOR_EN (Bit 30) c) INTERNAL_TX3_SIGNALS_MONITOR_EN (Bit 31) 3. Shortened the “ldoVmonScEn” field from 32 bits to 16 bits. a) Defeatured PA LDO short circuit and VMON circuit monitoring enable bits from this field. 4. Added a new 16 bit field “enMask2”. d) TX3_PHASE_SHIFTER_MONITOR_EN (Bit 0) e) TX_PHSHIFTER_DAC_MONITOR (Bit 1)
LOW	“digTempThreshMin” and “digTempThreshMax” fields in the RL_RF_TEMP_MON_CONF_SB API (rlTempMonConf_t) have been defeatured in xWR294x.
MEDIUM	Defeatured the production usage of “lpfCutoffBandEdgeDroopThresh”, ” lpfCutoffStopBandAttenThresh” fields in the RL_RF_RX_IFSTAGE_MON_CONF_SB API (rlRxIfStageMonConf_t).
LOW	Added support for providing the start frequency of the monitoring chirp – “monStartFreqConst” in RL_RF_TX0_BALLBREAK_MON_CONF_SB API - rlTxBallbreakMonConf_t.
MEDIUM	Defeatured the TX PS DAC monitor threshold field “txPhShiftDacMonThresh” in RL_RF_TXn_INT_ANA_SIGNALS_MON_CONF_SB APIs (rlTxIntAnaSignalsMonConf_t). Customers will have to use the new API - RL_RF_MON_TX_PHSHIFTER_DAC_CONF_SB to configure the TX PS DAC monitor in xWR294x.
LOW	The APLL clock (ClockPair 0) that is being monitored in the DCC clock monitor - RL_RF_DUAL_CLOCK_COMP_MON_CONF_SB API has been changed from 600MHz to 200MHz.
MEDIUM	Analog fault injection API - RL_RF_FAULT_INJECTION_CONF_SB has been modified in xWR294x as listed below. 1. Added a bit (3) to the “txGainDrop” field to enable TX gain fault for the 4^{th} TX. 2. Added a bit (6) to the “txPhInv” field to enable TX phase fault for the 4^{th} TX. 3. Modified the “supplyLdoFault” field to inject a fault using the TESTPATH LDO instead of the RX_LODIST_LDO. 4. Updated the “miscThreshFault” field to indicate that this fault is only injected in the GPADC_INTERNAL_SIGNALS_MONITOR. 5. Added a new “txPsDacFault” fault injection mechanism to cover the TX_PHSHIFTER_DAC monitor.
MEDIUM	Added a bit to enable/disable the 4^{th} TX in the “txChannelEn” field of the RL_RF_CHAN_CONF_SB (rlSetChannelConfig).
MEDIUM	Updated the “b2AdcOutFmt“ field in RL_RF_ADCOUT_CONF_SB API (rlAdcBitFormat_t) to only allow REAL ADC out format in xWR294x devices.
LOW	Added a bit in the “constBpmVal” field of the RL_RF_BPM_CHIRP_CONF_SB API

	(rlSetBpmChirpConfig) to allow setting the BPM value for the 4th TX.
HIGH	Updated the RL_RF_PROFILE_CONF_SB API (rlSetProfileConfig) for the following xWR294x changes. 1. "profileId" field now supports 7 profiles. 2. "pfVcoSelect" supported frequencies for VCO1 and VCO2 updated. 3. Updated the minimum valid value of the "rampEndTime" field to 2. 4. Added a byte in the field "txOutPowerBackoffCode" for controlling the backoff for the 4th TX. 5. Added a byte in the field "txPhaseShifter" for controlling the TX phase for the 4th TX. 6. Updated the valid range and recommended values for the field "freqSlopeConst". 7. Updated the valid range for the field "digOutSampleRate". 8. Updated the allowed values for the fields "hpfCornerFreq1" and "hpfCornerFreq2". 9. Modified the configuration values for the field "txCalibEnCfg" to account for the 4th TX. 10. Updated the "rxGain" field for the allowed range of values for the RX gain and RF gain targets. 11. Added a new field "miscFeatureEn" to control the enable/disable of miscellaneous features like HPF fast init and Multi TX enables during runtime calibration.
LOW	Updated the "txEnable" field in the RL_RF_CHIRP_CONF_SB API (rlSetChirpConfig) to add the control bit for the 4th TX.
MEDIUM	Updated the RL_RF_CONT_STREAMING_MODE_CONF_SB API (rlSetContModeConfig) for the following xWR294x changes. 1. Added a byte in the field "txOutPowerBackoffCode" for controlling the backoff for the 4th TX. 2. Added a byte in the field "txPhaseShifter" for controlling the TX phase for the 4th TX. 3. Updated the allowed values for the fields "hpfCornerFreq1" and "hpfCornerFreq2". 4. Updated the "rxGain" field for the allowed range of values for the RX gain and RF gain targets. 5. Updated the "vcoSelect" field for the valid range of the frequencies allowed for VCO1 and VCO2.
LOW	Modified the structure "rlRfTempData_t" with the following changes. DIGITAL temperature sensors are no longer read by the DFP. It can be read directly from the MSS using the MSS GPADC. 1. "tmpDig0Sens" now reports the 4th TX temperature sensor reading. 2. "tmpDig1Sens" field is now reserved.
LOW	Added a bit in the "bssAnaControl" field to enable/disable Inter burst APLL power save feature in the RL_RF_DEVICE_CFG_SB API (rlRfSetDeviceCfg).
LOW	Defeatured the "ldoBypassEnable" and "ioSupplyIndicator" field in the RL_RF_LDOBYPASS_SET_SB API (rlRfSetLdoBypassConfig).

LOW	Added a byte in the RL_RF_PERCHIRPPHASESHIFT_CONF_SB API (rlRfSetPhaseShiftConfig) for the 4th TX.
LOW	Updated the RL_RF_PALOOPBACK_CFG_SET_SB API (rlRfSetPALoopbackConfig). 1. Expanded the bitwidth and the valid values of the “paLoopbackFreq” field. 2. Added two new fields” comBufGainSel” and “paLbBufGainSel” to control the buffer gains in the PA loopback path.
LOW	Updated the allowed range of loopback frequencies in the “ifLoopbackFreq” field of the RL_RF_IFLOOPBACK_CFG_SET_SB API (rlRfSetIFLoopbackConfig).
LOW	De-featured the “progFiltFreqShift” field in the RL_RF_PROG_FILT_CONF_SET_SB API (rlRfSetProgFiltConfig)
MEDIUM	Updated the RL_RF_RADAR_MISC_CTL_SB API (rlRfSetMiscConfig) to add feature specific controls. 1. Added a bit (4) to the “miscCtl” field for CRD enable/disable. 2. Added the field “crdNSlopeMag” to control the ramp down slope magnitude. 3. Added the field “fastResetEndTime” to control the end of the fast reset pulse in the chirp’s idle time.
MEDIUM	Updated the RL_RF_CAL_MON_FREQ_LIMIT_SB API (rlRfSetCalMonFreqLimitConfig) 1. Added a new field “vco2RangeConfig” to select the range of VCO2. 2. Change the valid range of the “freqLimitHigh” field to be dependent on the the setting of the vco2RangeConfig field.
LOW	Updated the RL_RF_INIT_CALIB_CONF_SB API (rlRfInitCalibConfig) 1. Added a new field “txPowCalCfg” to control which TX chains have to be enabled when calibrating each TX chain in the TX power boot calibration. 2. De-featured the RX IQMM calibration.
MEDIUM	Updated the RL_RF_RUN_TIME_CALIB_CONF_TRIG_SB API (rlRfRunTimeCalibConfig) 1. Added TX phase shifter calibration run time calibration to the “oneTimeCalibEnMask” and “periodicCalibEnMask” fields. 2. Added a new temp index override enable bit (3) in the “CalTempIdxOverrideEn” field for the TX phase shifter calibration. 3. Added a new field “CalTempIdxTxPhase” for the override index.
LOW	Updated the interpretation of the “txGainTempLut” field in the below APIs. 1. rlAdvTxGainTempLutGet and rlAdvTxGainTempLutSet. 2. rlTxGainTempLutSet and rlTxGainTempLutGet
LOW	Updated the interpretation of the IF_GAIN_CODE and RF_GAIN_CODE of the “rxGainTempLut” field in the rlRxGainTempLutSet and rlRxGainTempLutGet APIs.
MEDIUM	Updated the RL_RF_TX_FREQ_PWR_LIMIT_SB API (rlRfTxFreqPwrLimitConfig) 1. Added a new field “vco2RangeConfig” to select the range of VCO2. 2. Change the valid range of the “freqLimitHighTx0/1/2” fields to be dependent on the the setting of the vco2RangeConfig field. 3. Added new fields “freqLimitLowTx3”, “freqLimitHighTx3” and “tx3PwrBackOff” for the frequency and TX power limit range for the 4th TX.

MEDIUM	Updates to RL_RF_LB_BURST_CFG_SET_SB API (rlSetLoopBckBurstCfg) 1. Updated the “loopbackSel” field to account for the new LO loopback options in xWR294x. 2. Added a byte in the “txBackoff” field for the 4th TX. 3. Updated the “rxGain” field for the modified RF gain target values. 4. Added a bit (3) in the “txEn” field for the 4th TX. 5. Added new bits (7:6) in the “bpmConfig” field for BPM value of the 4th TX. 6. Updated the IF loopback frequencies allowed in the “ifLbFreq” field. 7. Updates to the “ps1PgaIndx” and “ps2PgaIndx” fields with the updated buffer gain values for xWR294x. 8. Updates to the “psLbFreq” field with the supported frequency values for LO loopback. 9. Defeatured the “paLbFreq22xx” field and added new “paLbFreq29xx” field for setting the PA loopback frequency. 10. Added a new field “paLbBufGainSel” to select the buffer gain to be used in the PA loopback path.
LOW	Calibration data structure in the RL_RF_CAL_DATA_RD_WR_SB API (rlRfCalibDataRestore and rlRfCalibDataStore) is updated for xWR294x. The overall size hasn’t changed, but the interpretation of the data structure is different in xWR294x.
LOW	Added 2 bytes in the “digTxFreqShift” field of the RL_RF_DIG_COMP_EST_CTRL_SB API (rlRfInterRxGainPhaseConfig) for the 4th TX.
MEDIUM	Updated the “bssSysStatus“ field in the RL_RF_BOOTUP_BIST_STATUS_SB API structure - “rlRfBootStatusCfg_t” for the below additional boot time test status. 1. Bus safety test (Bit 27) 2. ECC aggregator test (Bit 28) 3. MPU test (Bit 29)
MEDIUM	Updated the RL_RF_PH_SHIFT_CAL_DATA_RD_WR_SB APIs (rlRfPhShiftCalibDataRestore and rlRfPhShiftCalibDataStore) 1. Updated the valid range of the “txIndex” field in “rlPhShiftCalibrationStore_t” and “rlPhShiftCalDataGetCfg_t” for the 4th TX. 2. Modified the mapping of the “observedPhShiftData[128]” field bytes with respect to the desired phase shift. 3.
MEDIUM	Updated the RL_RF_ADV_CHIRP_CFG_SET_SB API (rlSetAdvChirpConfig) 1. Added a new index - TX3_PHASE_SHIFTER for the “chirpParamIdx” field for the 4th TX phase shifter parameter. 2. Updated the CHIRP_PROFILE_SELECT parameters to allow 7 profiles. 3. Updated the CHIRP_TX_EN and CHIRP_BPM_VAL parameters to account for the 4th TX. 4. De-featured the “Reset at the end of the frame” option in the “resetMode” field when using Advanced Frame Configuration. 5.
MEDIUM	Updated the RL_RF_APLL_SYNTH_BW_CTL_SB API (rlRfApllSynthBwCtlConfig)

	1. Created fields "synthIcpTrimVco1", "synthRzTrimVco1", "synthIcpTrimVco2" and "synthRzTrimVco2" to be able to independently control the ICP and RZ trims for VCO1 and VCO2. 2. Modified the default values of "apllIcpTrim" and "apllRzTrimLpf" fields.
LOW	Updated the RL_RF_ADV_CHIRP_DYN_LUT_ADD_OFF_CONF_SB API (rlSetAdvChirpDynLUTAddrOffConfig) 1. Added a control bit (10) in the "addrMaskEn" field for enabling dynamic update of the LUT addr offset for the 4th TX phase shifter. 2. Added an extra 16 bits in the "lutAddressOffset" field for the 4th TX phase shifter setting.

6.2 Migration from AWR294x (DFP 2.4.9) to AWR254x/AWR2x4xP (DFP 2.4.18.1)

This section explains the steps to migrate from AWR294x (DFP 2.4.9) release to DFP 2.4.18.1 package.

Refer Section 3.5.3 mmWaveLink framework updates from DFP 2.4.11.0 onwards to latest DFP
Refer Section 3.5.8 mmWaveLink framework updates from DFP 2.4.15.0 onwards to latest DFP

7 Notes

Refer mmWave mcuplus SDK 4.x for integrating DFP with xWR254x/xWR2x4xP/xWR2x4xECO/xWR2x4xLC SDK.